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Reliable Waveform Relaxation Convergence Estimates for Atmosphere-Ice-Ocean Coupling

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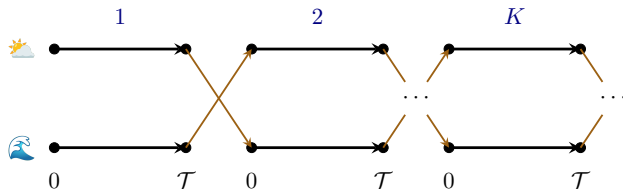
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Aim: Understand & improve coupling algorithms in climate codes w.r.t. accuracy & energy to solution.

Main Idea: View coupling as an iterative process, *waveform relaxation* (WR)

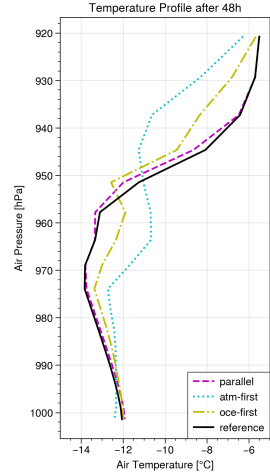


In each iteration $k = 1, 2, \dots$, the codes produce solutions U_A^k, U_O^k .

- Does this sequence converge? What is its limit?
- **How fast is convergence?** Can we accelerate it?

Why does this matter?

- Climate codes stop at $k = 1$
 - WR iteration error can be large
(cf. Marti et al. [1], V.S. et al. [2])
 - How can we reduce this error in practice?
- ⇒ Contribution to climate model development
e.g.: sea ice albedo fix from [2] part of EC-Earth 4.1.8



Question: How fast is WR convergence?

For (I) atmosphere-ocean, (II) atmosphere-ice, (III) atmosphere-ice-ocean coupling:

Continuous WR convergence factor ϱ

- Fully continuous analysis for linear coupled differential equations (Fourier techniques).

builds on: Blayo, Lemarié + collaborators¹

new results start at ✨

- Derive** ϱ for $\mathcal{T} = \infty$, **estimate** for $\mathcal{T} = 1$ h.

Observed WR convergence factor ϱ_{num}

- Tools:
 - CliMA Playground $\Rightarrow$ isolate discrete effects.
 - EC-Earth single column model (SCM) $\Rightarrow$ discrete effects + nonlinearities + physical processes.
- Measure** ϱ_{num} for $\mathcal{T} = 1$ h.

¹Clément [3], Lozano [4]

I: Atmosphere-Ocean Coupling

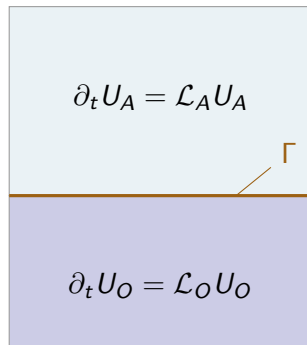


A Case of Domain Decomposition

- Two subdomains, interface = sea surface
- We focus on vertical turbulent thermodynamics: $U = T$
- Modeled as a diffusive process:

$$\partial_t T_j = \cdots + \alpha_j \partial_z^2 T_j, \quad j \in \{A, O\}$$

- Coupled heat equations, jump in material parameters



Waveform Relaxation

Compute for $k = 1, 2, \dots$

$$\begin{aligned}\partial_t T_A^k - \alpha_A \partial_z^2 T_A^k &= f_A \quad \text{on } (0, \mathcal{T}] \times (0, H_A), \\ T_A^k(0, z) &= \vartheta_A(z), \\ \partial_z T_A^k(t, H_A) &= g_A(t), \\ \text{Interface BC } \mathbf{1} &\text{ on } (0, \mathcal{T}] \times \Gamma,\end{aligned}\tag{1a}$$

and

$$\begin{aligned}\partial_t T_O^k - \alpha_O \partial_z^2 T_O^k &= f_O \quad \text{on } (0, \mathcal{T}] \times (-H_O, 0), \\ T_O^k(0, z) &= \vartheta_O(z), \\ \partial_z T_O^k(t, -H_O) &= g_O(t), \\ \text{Interface BC } \mathbf{2} &\text{ on } (0, \mathcal{T}] \times \Gamma.\end{aligned}\tag{1b}$$

The Interface Boundary Conditions

Known and well-studied: Dirichlet-Neumann (DN)

$$1. \quad T_A^k|_{\Gamma} = T_O^{k-1}|_{\Gamma} \quad 2. \quad k_O \partial_z T_O^k|_{\Gamma} = k_A \partial_z T_A^k|_{\Gamma}.$$

Standard in climate models: Bulk interface conditions

1. Flux $\propto$ temperature **jump** (law of the wall):

$$\begin{aligned} -k_A \partial_z T_A^k|_{\Gamma} &= C_{AO} \left(T_A^k|_{\Gamma} - T_O^{k-1}|_{\Gamma} \right) \\ \Leftrightarrow \left(k_A \partial_z T_A^k + C_{AO} T_A^k \right)|_{\Gamma} &= C_{AO} T_O^{k-1}|_{\Gamma} \end{aligned}$$

with $C_{AO} \propto (u_A - u_O)$.

2. Flux continuous across Γ :

$$k_O \partial_z T_O^k|_{\Gamma} = k_A \partial_z T_A^k|_{\Gamma}$$

Analysis Steps

1. Write coupling iteration (1) for the error in iteration k :

$$e^k(t, z) := T^k - T.$$

2. Apply Fourier transform in time, assuming $e^k = 0$ for $t \leq 0$, $T = \infty$:

$$\hat{e}^k(\omega, z) := \mathcal{F}\{e^k(t, z)\} \in \mathbb{C}.$$

3. Solve differential equations to obtain convergence factor:

$$\varrho(\omega) := \left| \frac{\hat{e}^k|_{\Gamma}}{\hat{e}^{k-1}|_{\Gamma}} \right|.$$

4. Estimate, cf. Gander and Halpern [5]:

$$\varrho_{\text{num}} \leq \max_{\omega \in \left[\frac{\pi}{T}, \frac{\pi}{\Delta t}\right]} \varrho(\omega).$$

Analysis Results

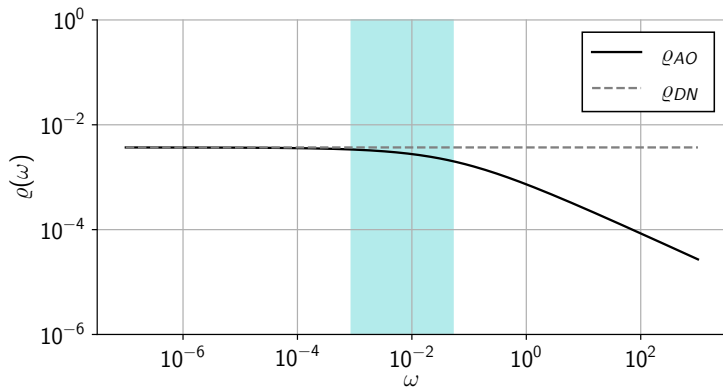
We obtain the convergence factor (cf. Clément [3]):

$$\varrho_{AO}(\omega) = \frac{k_A}{k_O} \sqrt{\frac{\alpha_O}{\alpha_A}} \left| \frac{\chi_A(\omega)}{\chi_O(\omega)} \right| \frac{1}{\left| 1 + \frac{k_A}{c_{AO}\sqrt{\alpha_A}} \sqrt{i\omega} \chi_A(\omega) \right|},$$
$$\varrho_{DN}(\omega) = \frac{k_A}{k_O} \sqrt{\frac{\alpha_O}{\alpha_A}} \left| \frac{\chi_A(\omega)}{\chi_O(\omega)} \right|,$$

with $\chi_j(\omega) = \tanh\left(H_j \sqrt{\frac{i\omega}{\alpha_j}}\right) \approx 1$.

Analysis Results: Continued

Assuming realistic material parameters for the atmosphere-ocean setting:



Shaded frequency band: $[\omega_{\min}, \omega_{\max}] = \left[\frac{\pi}{\Delta t_{\text{cpl}}}, \frac{\pi}{\Delta t_A} \right]$.

High-Frequency Decay ✨

We saw that

$$\varrho_{AO}(\omega) = \frac{\varrho_{DN}(\omega)}{|D_{AO}(\omega)|}, \quad D_{AO}(\omega) = 1 + \frac{k_A}{C_{AO}H_A} \sqrt{i\omega} \tanh\left(H_A \sqrt{\frac{i\omega}{\alpha_A}}\right) \in \mathbb{C}.$$

Consider real and imaginary parts separately

$$|D_{AO}|^2 = \operatorname{Re}(D_{AO})^2 + \operatorname{Im}(D_{AO})^2 = \left(1 + \frac{k_A}{C_{AO}H_A} f_1(\eta_A H_A)\right)^2 + \left(\frac{k_A}{C_{AO}H_A} f_2(\eta_A H_A)\right)^2,$$

with

$$\eta_A = \sqrt{\frac{\omega}{2\alpha_A}}, \quad f_1(x) := x \frac{\sinh(2x) - \sin(2x)}{\cosh(2x) + \cos(2x)}, \quad f_2(x) := x \frac{\sinh(2x) + \sin(2x)}{\cosh(2x) + \cos(2x)},$$

monotonically increasing for $\omega > 0 \Rightarrow \varrho_{AO}$ decreases $\Rightarrow$ **upper bound:** $\varrho_{AO}(\omega_{\min})$.

Putting on Modeling Glasses

In climate codes, material parameters nonlinearly depend on physical state!

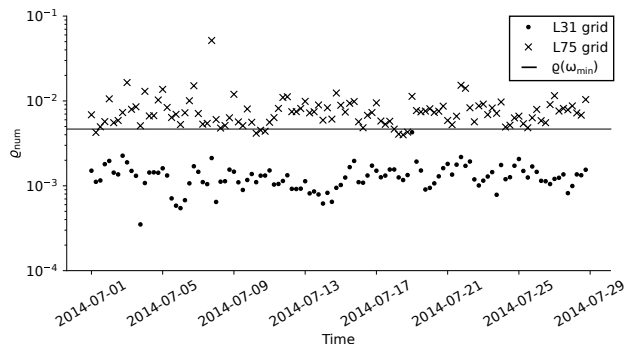
Most pronounced effect: $C_{AO} \propto (u_A - u_O)$.

How much can we expect ϱ_{AO} to vary in practice? Recall:

$$\varrho_{AO}(\omega_{\min}) \approx \varrho_{DN}(\omega_{\min}) \approx \frac{k_A}{k_O} \sqrt{\frac{\alpha_O}{\alpha_A}}.$$

$\Rightarrow$ Not so much, probably!

Enter: The EC-Earth SCM



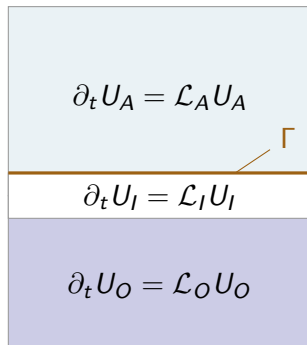
From analysis: $\varrho_{AO}(\omega_{\min}) \approx \frac{k_A}{k_O} \sqrt{\frac{\alpha_O}{\alpha_A}} = 3.6 \times 10^{-3} \Rightarrow$ Order of magnitude OK!
Coarser ocean grid reduces ϱ_{num} .

II: Atmosphere-Ice Coupling



Introducing Sea Ice [6]

- Ocean sees constant ice bottom temperature $T_I(t, -H_I)$
- ⇒ Decoupled! Only consider ice surface for analysis [4]
- Heat equation for ice temperature $T_I(t)$ ✨



Waveform Relaxation (reprise)

Compute for $k = 1, 2, \dots$

$$\begin{aligned}\partial_t T_A^k - \alpha_A \partial_z^2 T_A^k &= f_A \quad \text{on } (0, \mathcal{T}] \times (0, H_A), \\ T_A^k(0, z) &= \vartheta_A(z), \\ \partial_z T_A^k(t, H_A) &= g_A(t), \\ \text{IBC } \mathbf{1} &\text{ on } (0, \mathcal{T}] \times \Gamma,\end{aligned}$$

and

$$\begin{aligned}\partial_t T_I^k - \alpha_I \partial_z^2 T_I^k &= f_I \quad \text{on } (0, \mathcal{T}] \times (-H_I, 0), \\ T_I^k(0, z) &= \vartheta_I(z), \\ \partial_z T_I^k(t, -H_I) &= g_I(t), \\ \text{IBC } \mathbf{2} &\text{ on } (0, \mathcal{T}] \times \Gamma.\end{aligned}$$

The Interface Boundary Conditions (reprise)

Baseline: Dirichlet-Neumann (DN)

$$1. \quad T_A^k|_{\Gamma} = T_I^{k-1}|_{\Gamma} \quad 2. \quad k_I \partial_z T_I^k|_{\Gamma} = k_I \partial_z T_I^k|_{\Gamma}.$$

Standard in climate models: Bulk interface conditions + surface energy balance

1. Flux $\propto$ temperature jump (law of the wall):

$$\left(k_A \partial_z T_A^k + C_{AI} T_A^k \right) \Big|_{\Gamma} = C_{AI} T_I^{k-1} \Big|_{\Gamma}.$$

2. **Surface energy balance** over ice:

$$\left(k_I \partial_z T_I^k + (B + C_{AI}) T_I^k \right) \Big|_{\Gamma} = R + C_{AI} T_A^k \Big|_{\Gamma}.$$

$\Rightarrow$ Two Robin conditions!

Analysis Results: Ice-Covered Ocean

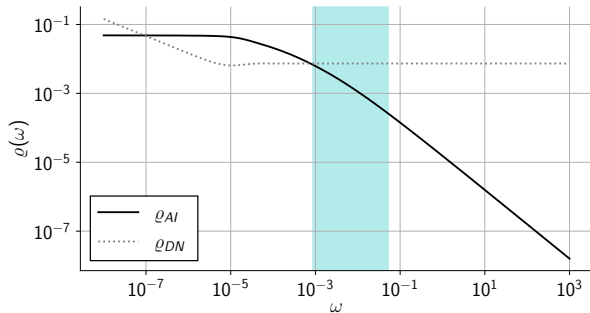
Resulting convergence factor:

$$\varrho_{AI}(\omega) = \frac{1}{\left| 1 + \frac{k_A}{C_{AI}} \sqrt{\frac{i\omega}{\alpha_A}} \chi_A(\omega) \right|} \frac{1}{\left| 1 + \frac{B}{C_{AI}} + \frac{k_I}{C_{AI}} \sqrt{\frac{i\omega}{\alpha_I}} \chi_I^{-1}(\omega) \right|},$$
$$\varrho_{DN}(\omega) = \frac{k_A}{k_I} \sqrt{\frac{\alpha_I}{\alpha_A}} \left| \frac{\chi_A(\omega)}{\chi_I(\omega)} \right|.$$

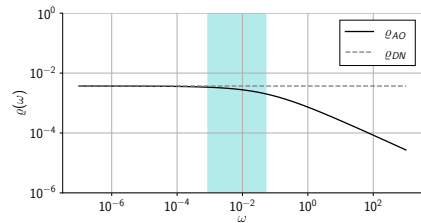
These have very different structure!

Analysis Results: Ice-Covered Ocean

With realistic material parameters, $H_I = 1$ m:



recall A-O coupling:



Shaded frequency band: $[\omega_{\min}, \omega_{\max}] = \left[\frac{\pi}{\Delta t_{\text{cpl}}}, \frac{\pi}{\Delta t_A} \right]$.

Putting on Modeling Glasses (reprise)

$$\varrho_{AI}(\omega) = \frac{1}{\left| 1 + \frac{k_A}{C_{AI}} \sqrt{\frac{i\omega}{\alpha_A}} \chi_A(\omega) \right|} \frac{1}{\left| 1 + \frac{B}{C_{AI}} + \frac{k_I}{C_{AI}} \sqrt{\frac{i\omega}{\alpha_I}} \chi_I^{-1}(\omega) \right|}$$

- Sea ice varies: ice thickness H_I , potentially snow-covered (affects k_I).
- Also: structure suggests stronger impact of $C_{AI} \propto (u_A - u_I)$ (compared to ϱ_{AO}).
- Let's look at the dependence of ϱ_{AI} on H_I , k_I , C_{AI} !

Changing k_I , C_{AI}

Given

$$\varrho_{AI}(\omega) = \frac{1}{|D_A(\omega)|} \frac{1}{|D_I(\omega)|}, \quad D_A(\omega), D_I(\omega) \in \mathbb{C}.$$

Same idea as previously: Split real and imaginary parts,

$$\begin{aligned} |D_A|^2 &= \left(1 + \frac{k_A}{C_{AI}H_A} f_1(\eta_A H_A)\right)^2 + \left(\frac{k_A}{C_{AI}H_A} f_2(\eta_A H_A)\right)^2, \\ |D_I|^2 &= \left(1 + \frac{B}{C_{AI}} + \frac{k_I}{C_{AI}H_I} f_3(\eta_I H_I)\right)^2 + \left(\frac{k_I}{C_{AI}H_I} f_4(\eta_I H_I)\right)^2. \end{aligned}$$

Note: Again, $|D_A|, |D_I|$ monotonically increasing in $\omega \Rightarrow$ upper bound $\varrho_{AI}(\omega_{\min})$.

- Larger $k_I \rightarrow$ lower ϱ_{AI} (faster convergence)
- Larger $C_{AI} \rightarrow$ higher ϱ_{AI} (slower convergence)
- How about H_I ?

Changing H_I

Ice thickness H_I only appears in $|D_I|$. Consider a fixed $\eta_I = \sqrt{\frac{\omega}{2\alpha_I}}$ and:

$$\begin{aligned}\operatorname{Re}(D_I) &= 1 + \frac{\varepsilon B}{C_{AI}} + \frac{k_I \eta_I}{C_{AI}} \frac{\sinh(2\eta_I H_I) + \sin(2\eta_I H_I)}{\cosh(2\eta_I H_I) - \cos(2\eta_I H_I)}, \\ \operatorname{Im}(D_I) &= \frac{k_I \eta_I}{C_{AI}} \frac{\sinh(2\eta_I H_I) - \sin(2\eta_I H_I)}{\cosh(2\eta_I H_I) - \cos(2\eta_I H_I)}.\end{aligned}$$

For the “hyperbolic-trigonometric” fractions, two regimes:

- large H_I : both behave like $\tanh(2\eta_I H_I) \approx 1$
 $\Rightarrow \varrho_{AI}$ independent of H_I ,
- (very) small H_I : $\operatorname{Im}(D_I) \rightarrow 0$, $\operatorname{Re}(D_I)$ behaves like $1/H_I$
 $\Rightarrow \varrho_{AI} \propto H_I$.

Enter: CliMA Playground

- Compare continuous analysis with discretized model
- Implementation “as we would do it in a climate model”:
isolate discrete effects!
- Code² based on CliMA³ & ClimaCoupler.jl
 - Implicit Euler in time
 - Centered FD in space
 - Multirate: e.g., $\Delta t_A < \Delta t_{\text{cpl}}$
 - Coupling variables time-averaged over Δt_{cpl}

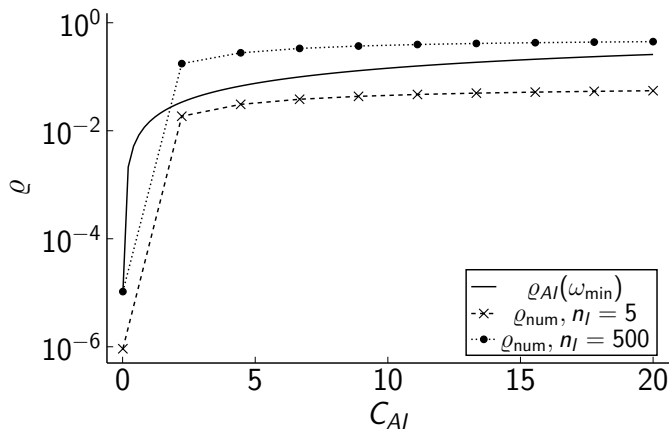
Discretization parameters

Δt_A	60 s
Δt_O	600 s
Δt_{cpl}	3600 s
$\mathcal{T}$	3600 s
Δz_A	1 m
Δz_O	1 m

²Code available at: <https://github.com/valentinaschuessler/clima-playground>

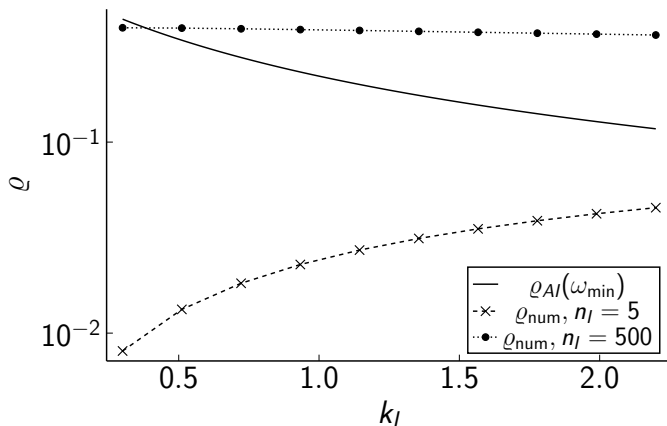
³<https://clima.caltech.edu/>

CliMA Playground (The Good?)



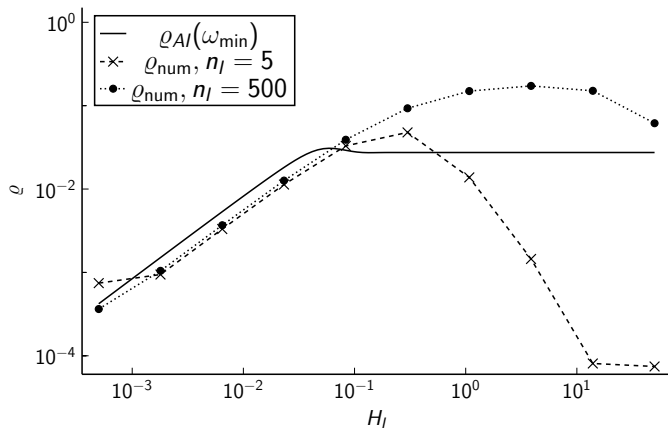
Discrete results agree with analysis: C_{AI} increase slows down convergence.

CliMA Playground (The Bad?)



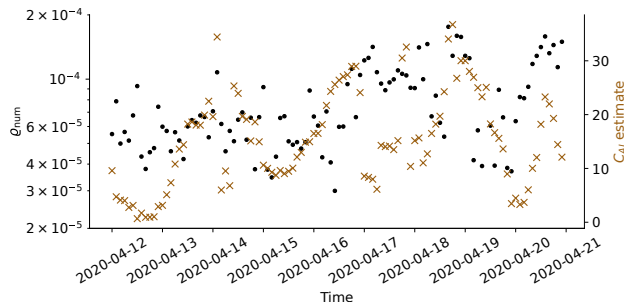
Discrete results do not agree with analysis, even under refinement!

CliMA Playground (The... Interesting?)



If $\Delta z_I = H_I/n_I$ large, Q_{num} is reduced.

Enter: The EC-Earth SCM



Roughly constant H_I , k_I ; C_{AI} varies significantly.

Very small convergence rates across the board! (Discrete H_I effect governs?)

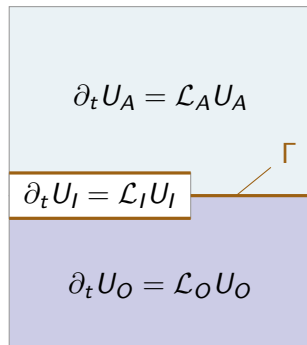
Differences hard to distinguish from noise.

III: Atmosphere-Ice-Ocean Coupling ✨



Partial Sea Ice Cover

- Sea ice covers fraction a_I of interface Γ
- Three subdomains!



Analysis Results: $a_I \in [0, 1]$

Linear combination of boundary conditions

$$k_A \partial_z T_A|_{\Gamma} = a_I C_{AI} (T_A|_{\Gamma} - T_I|_{\Gamma}) + (1 - a_I) C_{AO} (T_A|_{\Gamma} - T_O|_{\Gamma})$$

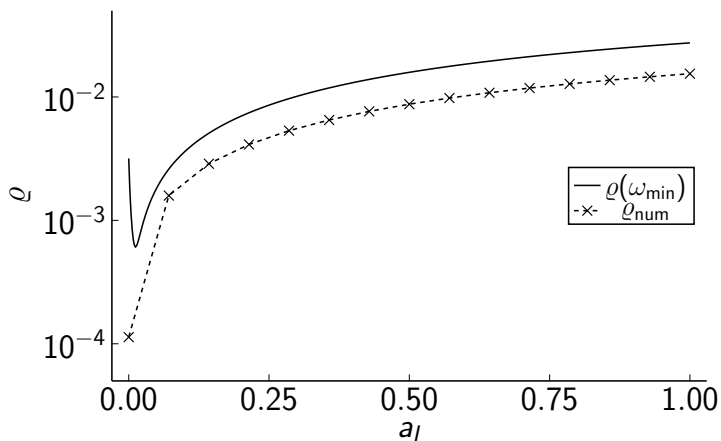
yields nonlinear combination of convergence factors:

$$\varrho(\omega) = \left| \frac{a_I \frac{C_{AI}^2}{B + C_{AI} + k_I \sqrt{\frac{i\omega}{\alpha_I}} \chi_I^{-1}(\omega)} - (1 - a_I)^2 \frac{k_A C_{AO} \sqrt{\frac{i\omega}{\alpha_A}} \chi_A(\omega)}{a_I C_{IO} + k_O \sqrt{\frac{i\omega}{\alpha_O}} \chi_O(\omega)} }{a_I C_{AI} + (1 - a_I) C_{AO} + k_A \sqrt{\frac{i\omega}{\alpha_A}} \chi_A(\omega)} \right|, \quad (3)$$

with

$$\lim_{a_I \rightarrow 0} \varrho = \varrho_{AO}, \quad \lim_{a_I \rightarrow 1} \varrho = \varrho_{AI}.$$

CliMA Playground: Varying Sea Ice Cover



Takeaways & Next Steps

- ✨ New analysis results (A-I, A-I-O coupling); direct comparison to realistic codes.
- WR convergence of atmosphere-ocean and atmosphere-ice coupling is **fast**.
 - A-O: “Good” conductivity/diffusivity ratio.
 - Bulk conditions, surface energy balance: Robin conditions, accelerate high frequency error decay.
 - Coarse grids seem to accelerate convergence further.
- In principle: Fast error reduction with few WR iterations possible.
- We observe **linear** WR convergence $\Rightarrow$ Fourier analysis gives a good first estimate.
- Discrete effects play an important role that we do not understand.
 - Remedy? Fully discrete analysis techniques from Monge and Birken [7] or Birken, Gander, Kotarsky (in prep.)
 - No results for Robin setting!



Děkuji!

 `valentinaschueller.github.io`

I code in public: `github.com/valentinaschueller`

References

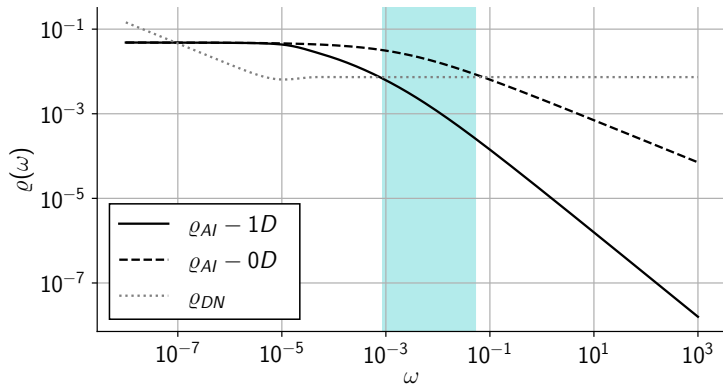
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- [7] A. Monge and P. Birken. “On the Convergence Rate of the Dirichlet–Neumann Iteration for Unsteady Thermal Fluid–Structure Interaction”. In: *Comput. Mech.* 62.3 (2018), pp. 525–541. DOI: 10.1007/s00466-017-1511-3.

Appendix



Analysis Results: Ice-Covered Ocean

Assuming realistic material parameters for the atmosphere-ice setting, $H_I = 1$ m:



Shaded frequency band: $[\omega_{\min}, \omega_{\max}] = \left[\frac{\pi}{\Delta t_{\text{cpl}}}, \frac{\pi}{\Delta t_A} \right]$.

Correlation between C_{AI} and wind speed

